



Arkansas State Standards Alignment

For Grades K-8

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About Zearn

Zearn is the 501(c)(3) nonprofit educational organization behind Zearn Math, the top-rated math learning platform used by 1 in 4 elementary-school students and by more than 1 million middle-school students nationwide. Zearn Math supports teachers with research-backed curriculum and digital lessons proven to double student academic growth across all levels of student proficiency. Zearn Math can be used flexibly to support a range of instructional needs including Tier 1 and 2 support, acceleration, intervention, tutoring, after school programming and summer programming.

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Zearn Math is aligned to Arkansas Mathematics Standards.

We are committed to ensuring that our high-quality instructional materials align to Arkansas Mathematics Standards to fully support Arkansas’s students and teachers. The tables on the following pages, organized by Arkansas Mathematics Standards, identify the alignment between each grade-level standard and Zearn Math lesson(s).

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Kindergarten

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
Kindergarten Standards		Lessons
Number & Place Value		
K.NPV.1	Count to 100 by ones and tens; count forward by ones from any given number up to 100.	Mission 5, Topic C, Lesson 13 Mission 5, Topic D, Lesson 15-18
K.NPV.2	Count a set of objects up to 20 using one-to-one correspondence, demonstrating that the last number stated indicates the number of objects in the set regardless of the arrangement.	Mission 1, Topic B, Lesson 5-6 Mission 1, Topic C, Lesson 7-11 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic E, Lesson 17-22 Mission 1, Topic F, Lesson 23-28 Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-37
K.NPV.3	Identify the position of objects in a set using ordinal numbers (first, second, third, etc.).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
K.NPV.4	Identify quickly a number of items in a set from 0 to 10 without counting.	<i>Digital Activities</i> <i>Zearn Math includes digital activities that allow for instantly recognizing a quantity in a structured arrangement.</i>
K.NPV.5	Read, write, and represent whole numbers from 0 to 20.	Mission 1, Topic D, Lesson 12-16 Mission 1, Topic E, Lesson 17-22 Mission 1, Topic F, Lesson 24, 26-28 Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-35, 37 Mission 4, Topic A, Lesson 1-6 Mission 4, Topic B, Lesson 7-12 Mission 4, Topic E, Lesson 25, 27 Mission 5, Topic A, Lesson 1-4 Mission 5, Topic B, Lesson 6, 9 Mission 5, Topic C, Lesson 10-13

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
Kindergarten Standards		Lessons
K.NPV.6	Show equivalent forms of whole numbers up to 20 as groups of tens and ones, using manipulatives and drawings.	Mission 5, Topic A, Lesson 2-5 Mission 5, Topic B, Lesson 6-9 Mission 5, Topic C, Lesson 10-13 Mission 5, Topic E, Lesson 20-21, 23
K.NPV.7	Use matching and counting strategies to compare the number of objects in one group to the number of objects in another group (0 to 10) using the terms greater than, less than, or equal.	Mission 3, Topic E, Lesson 17-19 Mission 3, Topic F, Lesson 20-24 Mission 3, Topic G, Lesson 25-28
K.NPV.8	Compare two whole numbers, using the terms greater than, less than, or equal.	Mission 3, Topic F, Lesson 20 Mission 3, Topic G, Lesson 25-28
Computation & Algebraic Reasoning		
K.CAR.1	Use objects, fingers, mental images, drawings, sounds, acting out situations, or verbal explanations to represent addition and subtraction from 0 to 10.	Mission 4, Topic C, Lesson 13-16 Mission 4, Topic D, Lesson 20-24 Mission 4, Topic F, Lesson 29-30 Mission 4, Topic G, Lesson 33 Mission 4, Topic H, Lesson 38
K.CAR.2	Use objects or drawings to decompose numbers less than or equal to 10 into pairs in more than one way, recording each decomposition.	Mission 1, Topic C, Lesson 9-11 Mission 1, Topic D, Lesson 14-15 Mission 4, Topic A, Lesson 2, 4-6 Mission 4, Topic B, Lesson 7-12 Mission 4, Topic D, Lesson 22-24 Mission 4, Topic E, Lesson 25-28 Mission 4, Topic F, Lesson 32
K.CAR.3	Use a drawing or equation to find the number that makes 10 when added to a given number.	Mission 4, Topic H, Lesson 39-40
K.CAR.4	Use manipulatives and various strategies to fluently add and subtract within 10.	Mission 4, Topic G, Lesson 34-36 Mission 4, Topic H, Lesson 37, 39 Mission 6, Topic A, Lesson 1-2 Mission 6, Topic B, Lesson 5
K.CAR.5	Solve real-world problems involving addition and subtraction within 10, using objects, drawings, or equations to represent the problem.	Mission 4, Topic C, Lesson 16-18 Mission 4, Topic D, Lesson 19, 21 Mission 4, Topic F, Lesson 31 Mission 4, Topic G, Lesson 34-36 Mission 4, Topic H, Lesson 37

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
Kindergarten Standards		Lessons
Geometry & Measurement		
K.GM.1	Describe the positions of objects and geometric shapes in the environment. Terms include: inside, outside, between, above, below, near, far, under, over, up, down, behind, in front of, next to, to the left of, and to the right of	Mission 2, Topic A, Lesson 5 Mission 2, Topic B, Lesson 8 Mission 2, Topic C, Lesson 10
K.GM.2	Name shapes correctly regardless of their orientation or overall size. Shapes include: squares, circles, triangles, rectangles, hexagons, cubes, cones, cylinders, and spheres	Mission 2, Topic A, Lesson 2-5 Mission 2, Topic B, Lesson 7-8 Mission 2, Topic C, Lesson 9-10
K.GM.3	Identify two-dimensional attributes of three-dimensional objects.	Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 6 Mission 2, Topic C, Lesson 9-10
K.GM.4	Analyze and sort a variety of two and three-dimensional shapes using informal language to describe their similarities, differences, and other attributes.	Mission 2, Topic A, Lesson 1-4 Mission 2, Topic B, Lesson 6-7 Mission 2, Topic C, Lesson 9-10 Mission 6, Topic A, Lesson 3
K.GM.5	Compose and draw shapes found in the world using objects (e.g., straws, toothpicks, clay balls).	Mission 6, Topic A, Lesson 1-3
K.GM.6	Make direct comparisons of the length, capacity, weight, and temperature of objects, recognizing which object is shorter/longer, lighter/heavier, warmer/cooler, or holds more.	Mission 3, Topic A, Lesson 1-3 Mission 3, Topic B, Lesson 4-7 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 13-15 Mission 3, Topic F, Lesson 20 Mission 3, Topic G, Lesson 28 Mission 3, Topic H, Lesson 29-32
K.GM.7	Understand concepts of time, recognizing that clocks and calendars are tools that measure time. Concepts of time include: morning, afternoon, evening, today, yesterday, tomorrow, day, week, month, and year	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
K.GM.8	Identify pennies and dimes by name and value.	Mission 4, Topic A, Lesson 6 Mission 6, Topic E, Lesson 20

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
Kindergarten Standards		Lessons
Data Analysis		
K.DA.1	Collect, sort, and organize data into two or three categories, using real-object graphs and picture graphs.	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 7 Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 7

1st Grade

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
1st Grade Standards		Lessons
Number & Place Value		
1.NPV.1	Count forward and back within 120 by ones and tens from any given whole number.	Mission 4, Topic A, Lesson 1-6 Mission 6, Topic B, Lesson 4, 7, 9
1.NPV.2	Skip count forward by multiples of fives within 120.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
1.NPV.3	Explain the place value of ones and tens in two-digit numbers, using concrete models, diagrams, numbers, or words.	Mission 2, Topic D, Lesson 26-29 Mission 4, Topic A, Lesson 1-4, 6 Mission 4, Topic F, Lesson 23 Mission 6, Topic B, Lesson 3-4, 8-9
1.NPV.4	Read, write, and represent whole numbers up to 120, using concrete models or drawings, word form, base ten numerals, and expanded form.	Mission 4, Topic A, Lesson 1-6 Mission 6, Topic B, Lesson 4, 7, 9
1.NPV.5	Use concrete models or drawings to subtract multiples of 10 from multiples of 10 (within the range of 10-90), relate the strategy to a written expression or equation, and explain the reasoning used to solve.	Mission 4, Topic C, Lesson 11 Mission 6, Topic C, Lesson 10
1.NPV.6	Use mental strategies to find 10 more or 10 less than a given two-digit number.	Mission 4, Topic A, Lesson 5-6 Mission 6, Topic B, Lesson 5
1.NPV.7	Compare two two-digit numbers using symbols ($<$, $=$, $>$) based on the value of tens and ones in the given numbers.	Mission 4, Topic B, Lesson 7-10 Mission 6, Topic B, Lesson 6
1.NPV.8	Partition circles and rectangles into two and four equal shares, describing the shares using the words halves, fourths, and quarters; understand that decomposing into more equal pieces creates smaller pieces.	Mission 5, Topic C, Lesson 7-9 Mission 5, Topic D, Lesson 10
Computation & Algebraic Reasoning		

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
1st Grade Standards		Lessons
1.CAR.1	Add and subtract fluently within 10 with mastery by the end of first grade	Mission 1, Topic C, Lesson 9-13 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic E, Lesson 17-20 Mission 1, Topic F, Lesson 21-24 Mission 1, Topic G, Lesson 26-27 Mission 1, Topic I, Lesson 33-37 Mission 1, Topic J, Lesson 38-39 Mission 2, Topic A, Lesson 1-11 Mission 2, Topic B, Lesson 12-21 Mission 2, Topic C, Lesson 25 Mission 2, Topic D, Lesson 27-29
1.CAR.2	Use computational fluency to add and subtract within 20 using manipulatives and/or a variety of strategies.	Mission 1, Topic C, Lesson 9-13 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic E, Lesson 17-20 Mission 1, Topic F, Lesson 21-24 Mission 1, Topic G, Lesson 26-27 Mission 1, Topic I, Lesson 33-37 Mission 1, Topic J, Lesson 38-39 Mission 2, Topic A, Lesson 1-11 Mission 2, Topic B, Lesson 12-21 Mission 2, Topic C, Lesson 25 Mission 2, Topic D, Lesson 27-29
1.CAR.3	Apply properties of operations to add and subtract within 20.	Mission 1, Topic E, Lesson 19-20 Mission 1, Topic F, Lesson 21-24 Mission 2, Topic A, Lesson 2, 6, 10-11
1.CAR.4	Use concrete models or drawings to add within 100, including a two-digit number and a one-digit number as well as a two-digit number and a multiple of ten; relate strategy used to a written expression or equation and explain reasoning.	Mission 4, Topic C, Lesson 11-12 Mission 4, Topic D, Lesson 13-18 Mission 4, Topic F, Lesson 23-29 Mission 6, Topic C, Lesson 10-17 Mission 6, Topic D, Lesson 18-19
1.CAR.5	Demonstrate the relationship between addition and subtraction by solving problems, using an inverse operation.	Mission 1, Topic C, Lesson 9-13 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic E, Lesson 17-20 Mission 1, Topic F, Lesson 21-24 Mission 1, Topic G, Lesson 26-27 Mission 1, Topic I, Lesson 33-37 Mission 1, Topic J, Lesson 38-39 Mission 2, Topic A, Lesson 1-11 Mission 2, Topic B, Lesson 12-21

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
1st Grade Standards		Lessons
		Mission 2, Topic C, Lesson 25 Mission 2, Topic D, Lesson 27-29
1.CAR.6	Solve real-world problems involving addition and subtraction within 20. Problem types include: adding to, taking from, putting together, taking apart, and comparing with unknowns present throughout the addition and subtraction problem.	Mission 1, Topic B, Lesson 4-8 Mission 1, Topic C, Lesson 9-13 Mission 1, Topic G, Lesson 25 Mission 1, Topic H, Lesson 28-32 Mission 2, Topic A, Lesson 3-5, 7-9, 11 Mission 2, Topic B, Lesson 12-13, 17-18, 21 Mission 2, Topic C, Lesson 22-24 Mission 2, Topic D, Lesson 27-29 Mission 3, Topic D, Lesson 10-13 Mission 4, Topic E, Lesson 19-22 Mission 6, Topic A, Lesson 1-2 Mission 6, Topic F, Lesson 25-27
1.CAR.7	Solve real-world problems involving addition of three whole numbers whose sum is less than or equal to 20.	Mission 2, Topic A, Lesson 1
1.CAR.8	Apply understanding of the equal sign to determine if equations involving addition and subtraction are true or false.	Mission 1, Topic E, Lesson 17-19 Mission 2, Topic A, Lesson 6, 10 Mission 2, Topic B, Lesson 20 Mission 2, Topic C, Lesson 25
1.CAR.9	Determine the unknown whole number in an addition or subtraction equation relating three whole numbers.	Mission 1, Topic D, Lesson 16 Mission 1, Topic H, Lesson 29-30, 32 Mission 2, Topic A, Lesson 10 Mission 2, Topic B, Lesson 20
Geometry & Measurement		
1.GM.1	Understand the difference between defining attributes (e.DA., triangles are closed and three-sided shapes) and non-defining attributes (e.DA., color, orientation, overall size), using that understanding to build and draw shapes that exhibit defining attributes.	Mission 5, Topic A, Lesson 1-3
1.GM.2	Create a composite shape using two-dimensional or three-dimensional shapes.	Mission 5, Topic B, Lesson 4-6

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
1st Grade Standards		Lessons
	- Two-dimensional include: rectangle, square, trapezoid, triangle, hexagon, half circle, and quarter circle - Three-dimensional include: cube, rectangular prism, cone, and cylinder	
1.GM.3	Express the length of an object as a whole number of units by laying multiple copies of a shorter object end to-end, understanding that the length of one object is equal to the number of same-size units that span the object with no gaps or overlaps.	Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 7-9
1.GM.4	Order three objects by their length, indirectly comparing the lengths of two objects by using a third object.	Mission 3, Topic A, Lesson 1-3 Mission 3, Topic B, Lesson 5-6 Mission 3, Topic C, Lesson 9
1.GM.5	Tell and write time to the nearest hour and half hour using analog clocks; understand how to read hours and minutes using digital clocks.	Mission 5, Topic D, Lesson 10-13
1.GM.6	Identify coins by name and value, including penny, nickel, dime, and quarter.	Mission 5, Topic D, Lesson 10-13
1.GM.7	Count collections of like coins including pennies, nickels, and dimes to determine their total value up to 100 cents.	Mission 6, Topic E, Lesson 20-24
Data Analysis		
1.DA.1	Organize, represent, and interpret data with up to three categories (e.g., tally tables, picture graphs, bar graphs).	Mission 3, Topic D, Lesson 10-13
1.DA.2	Ask and answer questions about the total number represented such as how many in each category and how many more or less in one category compared to another.	Mission 3, Topic D, Lesson 10-13

2nd Grade

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
2nd Grade Standards		Lessons
Number & Place Value		
2.NPV.1	Count within 1,000 forwards and backwards by ones, tens, and hundreds from any given number.	Mission 3, Topic B, Lesson 2-3 Mission 3, Topic C, Lesson 4 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 12-15 Mission 3, Topic G, Lesson 19-21 Mission 7, Topic E, Lesson 21 Mission 8, Topic D, Lesson 14
2.NPV.2	Identify the value of hundreds, tens, and ones place in a three-digit number.	Mission 3, Topic A, Lesson 1 Mission 3, Topic B, Lesson 2-3 Mission 3, Topic C, Lesson 4-7 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 11-15
2.NPV.3	Read, write, and represent whole numbers up to 1,000 using concrete models or drawings, number names, and a variety of expanded forms.	Mission 3, Topic C, Lesson 4-7 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 13-14
2.NPV.4	Mentally add 10 or 100 to a given number in the range of 100-900 and mentally subtract 10 or 100 from a given number in the range of 100-900.	Mission 3, Topic G, Lesson 19-20 Mission 4, Topic D, Lesson 17 Mission 5, Topic A, Lesson 1-2
2.NPV.5	Compare two three-digit numbers using symbols ($<$, $=$, $>$) based on the value of hundreds, tens, and ones in the given numbers.	Mission 3, Topic F, Lesson 16-18
2.NPV.6	Partition circles and rectangles into two, three, or four equal shares, describing the shares using the words halves, thirds, and fourths (or quarters).	Mission 8, Topic B, Lesson 7-8 Mission 8, Topic C, Lesson 9-12 Mission 8, Topic D, Lesson 13
2.NPV.7	Recognize that equal shares of identical wholes need not have the same shape.	Mission 8, Topic B, Lesson 7-8 Mission 8, Topic C, Lesson 9-12 Mission 8, Topic D, Lesson 13

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
2nd Grade Standards		Lessons
Computation & Algebraic Reasoning		
2.CAR.1	Use mental strategies to fluently add and subtract within 20 with mastery by the end of second grade	Mission 1, Topic A, Lesson 1-2 Mission 1, Topic B, Lesson 4-7
2.CAR.2	Use computational fluency to add and subtract within 100 using strategies based on place value, properties of operations, or the relationship between addition and subtraction.	Mission 1, Topic B, Lesson 3, 5-6, 8 Mission 4, Topic A, Lesson 2-4 Mission 4, Topic B, Lesson 6-8, 10 Mission 4, Topic C, Lesson 11-13 Mission 7, Topic B, Lesson 7-8, 11-13
2.CAR.3	Add up to four two-digit numbers with sums not exceeding 100 using strategies based on place value and properties of operations.	Mission 4, Topic D, Lesson 22
2.CAR.4	Use a number line to solve addition and subtraction problems within 100.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
2.CAR.5	Use addition to find the total number of objects arranged in rectangular arrays with up to 5 rows and up to 5 columns; write an equation to express the total as a sum of equal addends.	Mission 6, Topic A, Lesson 1-4 Mission 6, Topic B, Lesson 5-9 Mission 6, Topic C, Lesson 10-15
2.CAR.6	Use concrete models, drawings, or equations to solve addition and subtraction problems within 1000	Mission 4, Topic B, Lesson 9 Mission 4, Topic C, Lesson 14-15 Mission 4, Topic D, Lesson 17-21 Mission 4, Topic E, Lesson 23-28 Mission 4, Topic F, Lesson 29-30 Mission 5, Topic A, Lesson 1-7 Mission 5, Topic B, Lesson 8-12 Mission 5, Topic C, Lesson 13-18 Mission 5, Topic D, Lesson 19-20
2.CAR.7	Solve one and two-step real-world problems involving addition and subtraction within 100 in situations of adding to, taking from, putting together, taking apart, and comparing unknowns in all positions.	Mission 1, Topic B, Lesson 3, 5-6, 8 Mission 4, Topic A, Lesson 1-5 Mission 4, Topic B, Lesson 6-10 Mission 4, Topic C, Lesson 11-14, 16 Mission 4, Topic F, Lesson 31 Mission 7, Topic B, Lesson 7-8, 12-13

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
2nd Grade Standards		Lessons
2.CAR.8	Determine whether a group of objects up to 20 has an odd or even number of members; write an equation to express an even number as a sum of two equal addends.	Mission 6, Topic D, Lesson 17-19
Geometry & Measurement		
2.GM.1	Identify, describe, and draw two-dimensional shapes. Shapes include: triangles, regular pentagons, regular hexagons, and quadrilaterals (square, rectangle, trapezoid, parallelogram, rhombus)	Mission 8, Topic A, Lesson 1-5 Mission 8, Topic B, Lesson 6
2.GM.2	Identify and describe three-dimensional shapes based on the shape, number of faces, number of edges, and number of vertices. Shapes include: rectangular prisms, cubes, and square-based pyramids	Mission 8, Topic B, Lesson 6
2.GM.3	Select appropriate measurement tools to estimate and measure the length of an object to the nearest whole inch or whole centimeters.	Mission 2, Topic A, Lesson 1-3 Mission 2, Topic B, Lesson 4-5 Mission 2, Topic C, Lesson 6 Mission 7, Topic C, Lesson 14-17 <i>Zearn Math includes lessons estimating and measuring length, extending beyond inches and centimeters. Teachers might need to adjust their instruction to ensure proper alignment.</i>
2.GM.4	Demonstrate how the length of an object does not change, regardless of the units used to measure it, by measuring the length of an object twice; use two different length units, describing how the two measurements relate to the size of the chosen unit.	Mission 2, Topic C, Lesson 7 Mission 2, Topic D, Lesson 9 Mission 7, Topic D, Lesson 18
2.GM.5	Measure to determine how much longer or shorter one object is than another, expressing the length difference in terms of a standard length whole unit.	Mission 2, Topic C, Lesson 6-7 Mission 2, Topic D, Lesson 9 Mission 7, Topic D, Lesson 19
2.GM.6	Solve real-world problems involving lengths of the same units, using addition and subtraction within 100.	Mission 2, Topic D, Lesson 8-10 Mission 7, Topic E, Lesson 20, 22

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
2nd Grade Standards		Lessons
2.GM.7	Solve real-world and mathematical problems to find the perimeter of polygons.	Grade 3, Mission 7, Topic C, Lesson 10-17 Grade 3, Mission 7, Topic D, Lesson 18-22 Grade 3, Mission 7, Topic E, Lesson 23-30
2.GM.8	Partition a rectangle into rows and columns of same-size squares, counting the total number of squares to find the area.	Mission 6, Topic C, Lesson 14-15
2.GM.9	Using an analog clock, tell and write time to the nearest five minutes using colon notation and indicate a.m. or p.m.	Mission 8, Topic D, Lesson 13-15
2.GM.10	Describe relationships of time. • Times include: seconds in a minute; minutes in an hour; hours in a day; days in a week; and days, weeks, and months in a year	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
2.GM.11	Solve real-world problems involving addition and subtraction of time intervals in half hours or hours.	Mission 8, Topic D, Lesson 16
2.GM.12	Count collections of mixed coins and solve real-world problems involving quarters, dimes, nickels, and pennies within 99¢ and whole dollar amounts.	Mission 3, Topic D, Lesson 10 Mission 7, Topic B, Lesson 6-10, 13
Data Analysis		
2.DA.1	Use bar graphs, picture graphs, and line plots to organize and represent data, interpreting data with up to four categories.	Mission 7, Topic A, Lesson 1-5
2.DA.2	Ask and answer simple put together, take apart, and compare problems, using information presented in the bar graphs, picture graphs, and line plots.	Mission 7, Topic A, Lesson 1-5

3rd Grade

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
3rd Grade Standards		Lessons
Number & Place Value		
3.NPV.1	Round four-digit whole numbers to the nearest 10 or 100, using place value understanding.	Mission 2, Topic C, Lesson 12-14 Mission 2, Topic D, Lesson 17 Mission 2, Topic E, Lesson 20-21
3.NPV.2	Identify the value of thousands, hundreds, tens, and ones place in a four-digit number.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
3.NPV.3	Read and write whole numbers up to 10,000, using base ten numerals, word form, and a variety of expanded forms.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
3.NPV.4	Compare two four-digit numbers using symbols ($<$, $=$, $>$) based on the value of thousands, hundreds, tens, and ones in the given numbers.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
3.NPV.5	Compare two fractions with the same numerator or denominator by reasoning about their size based on the same whole; use symbols ($<$, $=$, $>$) and justify the conclusion using visual fraction models, concrete objects, or words.	Mission 5, Topic C, Lesson 10-11, 13 Mission 5, Topic D, Lesson 18-19 Mission 5, Topic F, Lesson 28-29
3.NPV.6	Identify fractions as parts of a whole and parts of a collection or set. Fractions include: denominators 2, 3, 4, 6, and 8	Mission 5, Topic B, Lesson 5-9 Mission 5, Topic C, Lesson 10, 12-13 Mission 5, Topic E, Lesson 20, 22, 24-25, 27 Mission 5, Topic F, Lesson 28-29
3.NPV.7	Partition squares, regular hexagons, and equilateral triangles into parts with equal shares, explaining the shares of each part as a unit fraction of the whole. Fractions include: denominators 2, 3, 4, 6, and 8	Mission 5, Topic A, Lesson 1-4 Mission 5, Topic B, Lesson 5-9 Mission 5, Topic C, Lesson 10-13 Mission 5, Topic E, Lesson 24-25, 27 Mission 5, Topic F, Lesson 28-29

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
3rd Grade Standards		Lessons
3.NPV.8	Identify and represent a unit fraction as a number on the number line. Fractions include: denominators 2, 3, 4, 6, and 8	Mission 5, Topic D, Lesson 14, 19 Mission 5, Topic E, Lesson 21, 23, 26
3.NPV.9	Identify and represent a non-unit fraction as a number on the number line, including fractions greater than one. Fractions include: denominators 2, 3, 4, 6, and 8	Mission 5, Topic D, Lesson 14-17, 19 Mission 5, Topic E, Lesson 21, 23, 26
3.NPV.10	Decompose and compose a non-unit fraction a/b as the quantity formed by the sum of unit fractions. Fractions include: denominators 2, 3, 4, 6, and 8	Grade 4, Mission 5, Topic A, Lesson 1-6 Grade 4, Mission 5, Topic D, Lesson 16-18 Grade 4, Mission 5, Topic E, Lesson 22, 24-25 Grade 4, Mission 5, Topic F, Lesson 30, 32
3.NPV.11	Use number lines and visual models to recognize and generate equivalent fractions, explaining how they are equivalent in real-world and mathematical situations. Fractions include: denominators 2, 3, 4, 6, and 8	Mission 5, Topic E, Lesson 20-27
Computation & Algebraic Reasoning		
3.CAR.1	Use computational fluency to add and subtract three-digit whole numbers, using strategies and algorithms based on place value, properties of operations, and/or the relationship between addition and subtraction.	Mission 2, Topic A, Lesson 4-5 Mission 2, Topic B, Lesson 10-11 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21
3.CAR.2	Use basic fact fluency to multiply and divide whole numbers with mastery by the end of third grade. - Knowing all products with factors up to and including 12 and the corresponding division facts from the products with factors up to and including 12. - Using strategies such as the relationship between multiplication and division (e.g., Knowing that $8 \cdot 5 = 40$, one knows $40 \div 5 = 8$) or properties of operations.	Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14, 16 Mission 3, Topic B, Lesson 4-5 Mission 3, Topic C, Lesson 10 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
3rd Grade Standards		Lessons
3.CAR.3	Apply properties of operations as strategies to multiply and divide. Properties include: Distributive, Commutative, and Associative Properties of Multiplication	Mission 1, Topic C, Lesson 7-10 Mission 1, Topic E, Lesson 15-16 Mission 1, Topic F, Lesson 18-19 Mission 3, Topic A, Lesson 1-2 Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 9-10 Mission 3, Topic D, Lesson 12-14 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 20
3.CAR.4	Use strategies to multiply one-digit numbers by multiples of 10 ranging from 10-90; strategies are based on place value and properties of operations (e.g., $9 \cdot 80$, $5 \cdot 60$).	Mission 3, Topic F, Lesson 19-20
3.CAR.5	Identify arithmetic patterns including, but not limited to, patterns in an addition or multiplication table, explaining use of properties of operations appropriate to the pattern.	Mission 3, Topic A, Lesson 1 Mission 3, Topic D, Lesson 13-14 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19
3.CAR.6	Solve real-world problems using multiplication and division within 100 involving equal groups, arrays, partitive and measurement division.	Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 8-9 Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14-17 Mission 1, Topic F, Lesson 18 Mission 3, Topic A, Lesson 2-3 Mission 3, Topic B, Lesson 7 Mission 3, Topic C, Lesson 11 Mission 3, Topic D, Lesson 15
3.CAR.7	Solve two-step real-world situations using addition, subtraction, multiplication, and division, representing these problems using equations with a symbol standing for an unknown quantity.	Mission 1, Topic F, Lesson 20-21 Mission 3, Topic E, Lesson 18 Mission 3, Topic F, Lesson 21 Mission 7, Topic A, Lesson 1-3
3.CAR.8	Determine the unknown whole number in a multiplication or division equation relating three whole numbers.	Mission 1, Topic B, Lesson 6 Mission 1, Topic C, Lesson 8 Mission 1, Topic D, Lesson 12 Mission 1, Topic E, Lesson 17 Mission 3, Topic A, Lesson 3 Mission 3, Topic D, Lesson 13 Mission 3, Topic E, Lesson 16
3.CAR.9	Understand division as an unknown-factor problem.	Mission 1, Topic B, Lesson 6

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
3rd Grade Standards		Lessons
		Mission 1, Topic D, Lesson 11 Mission 1, Topic E, Lesson 17 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic E, Lesson 16
Geometry & Measurement		
3.GM.1	Understand that quadrilaterals in different categories may share attributes; those attributes (e.g., four equivalent sides) can define a larger category (e.g., quadrilaterals) or subcategory (e.g., rhombus and square).	Mission 7, Topic B, Lesson 4-9
3.GM.2	Identify perpendicular and parallel lines, as well as right angles in two-dimensional shapes and real-world surroundings or objects.	Mission 7, Topic B, Lesson 4-9
3.GM.3	Recognize rhombuses, rectangles, and squares as examples of quadrilaterals, identifying and/or drawing examples of quadrilaterals that do not belong to any of these subcategories.	Mission 7, Topic B, Lesson 4-9
3.GM.4	Measure lengths of objects to the nearest half and quarter inch, using a ruler.	Mission 6, Topic B, Lesson 5-9 Mission 7, Topic D, Lesson 19, 22 Mission 7, Topic E, Lesson 26
3.GM.5	Describe area as the number of unit squares that cover a plane figure without gaps and overlaps.	Mission 4, Topic A, Lesson 1-4 Mission 4, Topic B, Lesson 5-7 Mission 4, Topic D, Lesson 13
3.GM.6	Find the area of a rectangle with whole number side lengths by modeling with unit squares and multiplying the side lengths to show the results are the same.	Mission 4, Topic A, Lesson 4 Mission 4, Topic B, Lesson 5-8
3.GM.7	Multiply side lengths to find areas of rectangles with whole number side lengths in the context of solving real world and mathematical problems.	Mission 4, Topic B, Lesson 5-8 Mission 4, Topic C, Lesson 9-11 Mission 4, Topic D, Lesson 12-16 Mission 7, Topic E, Lesson 28-30
3.GM.8	Measure and estimate liquid volumes and masses of objects using standard units.	Mission 2, Topic B, Lesson 6-7, 9-10 Mission 2, Topic C, Lesson 12

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
3rd Grade Standards		Lessons
3.GM.9	Solve one-step real-world problems involving liquid volumes and masses of objects in the same units, using all four operations.	Mission 2, Topic C, Lesson 8, 11 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21
3.GM.10	Tell and write time to the nearest minute, using analog clocks.	Mission 2, Topic A, Lesson 1-5
3.GM.11	Solve word problems involving addition and subtraction of time intervals in minutes.	Mission 2, Topic A, Lesson 4-5 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18, 21
Data Analysis		
3.DA.1	Represent a data set with multiple categories, using a scaled picture graph, scaled bar graph, and a line plot.	Mission 6, Topic A, Lesson 1-4
3.DA.2	Solve one and two-step problems, using categorical data represented with a scaled picture graph, scaled bar graph, and a line plot.	Mission 6, Topic A, Lesson 1-4

4th Grade

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
4th Grade Standards		Lessons
Number & Place Value		
4.NPV.1	Recognize that a digit in a given place represents ten times what it represents in the place to its right.	Mission 1, Topic A, Lesson 2-3
4.NPV.2	Read and write whole numbers up to 1,000,000 using base ten numerals, word form, and a variety of expanded forms.	Mission 1, Topic A, Lesson 3-4 Mission 1, Topic B, Lesson 5-6
4.NPV.3	Use place value understanding to round five-digit and six-digit whole numbers to any place.	Mission 1, Topic C, Lesson 7-10 Mission 1, Topic D, Lesson 12 Mission 1, Topic E, Lesson 16-17 Mission 1, Topic F, Lesson 19
4.NPV.4	Compare two five-digit whole numbers and six-digit whole numbers, using symbols ($<$, $=$, $>$) to record the results of comparisons.	Mission 1, Topic B, Lesson 5-6
4.NPV.5	Compare two fractions with different numerators and different denominators using symbols ($<$, $=$, $>$) to record the results of comparisons (e.g., by creating common denominators or numerators or by comparing to a benchmark of 0 , $\frac{1}{2}$, 1).	Mission 5, Topic C, Lesson 12-15 Mission 5, Topic E, Lesson 26-27
4.NPV.6	Compare two decimals to the hundredths place, using symbols ($<$, $=$, $>$) to record the results of comparisons.	Mission 6, Topic C, Lesson 9-11
4.NPV.7	Decompose fractions, including fractions greater than one and mixed numbers, into unit fractions, using concrete models, drawings, and/or the number line. Fractions include denominators 2, 3, 4, 5, 6, 8, 10, 12, and 100.	Mission 5, Topic A, Lesson 1-6 Mission 5, Topic D, Lesson 16-18 Mission 5, Topic E, Lesson 22, 24-25 Mission 5, Topic F, Lesson 30, 32
4.NPV.8	Explain why a fraction $\frac{a}{b}$ is equivalent to a fraction $\frac{n \cdot a}{n \cdot b}$, using visual fraction models, generating equivalent fractions using the principle $\frac{a}{b} = \frac{n \cdot a}{n \cdot b}$. Fractions include denominators 2, 3, 4, 5, 6, 8, 10, 12, and 100.	Mission 5, Topic B, Lesson 7-11 Mission 5, Topic C, Lesson 14-15 Mission 5, Topic D, Lesson 21 Mission 5, Topic E, Lesson 27 Mission 6, Topic B, Lesson 5 Mission 6, Topic D, Lesson 12

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
4th Grade Standards		Lessons
4.NPV.9	Add two fractions with denominators of 10 and 100 by expressing the denominator of 10 as an equivalent fraction with a denominator of 100.	Mission 6, Topic B, Lesson 4 Mission 6, Topic D, Lesson 12-14
4.NPV.10	Apply decimal notation for fractions with denominators 10 or 100.	Mission 6, Topic A, Lesson 1-3 Mission 6, Topic B, Lesson 4-6 Mission 6, Topic C, Lesson 11 Mission 6, Topic D, Lesson 12-14 Mission 6, Topic E, Lesson 15
Computation & Algebraic Reasoning		
4.CAR.1	Find the factor pairs for a given number in the range of 1-100, identifying whether a number is prime or composite; determine whether a given whole number in the range of 1-100 is a multiple of a given one-digit number.	Mission 3, Topic F, Lesson 22-25
4.CAR.2	Use computational fluency to add and subtract whole numbers up to 1,000,000 by using strategies and algorithms, including the standard algorithm, with mastery by the end of fourth grade.	Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14-17 Mission 1, Topic F, Lesson 18-19
4.CAR.3	Use strategies based on place value and the properties of operations to multiply four-digit by one-digit whole numbers and two two-digit whole numbers.	Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 7-11 Mission 3, Topic H, Lesson 34-38
4.CAR.4	Use strategies based on place value, the properties of operations, and the relationship between multiplication and division to divide whole numbers with four-digits by one-digit divisors; quotients should be with and without whole number remainders.	Mission 3, Topic E, Lesson 14-21 Mission 3, Topic G, Lesson 26-33
4.CAR.5	Add and subtract fractions, including mixed numbers, with like denominators, using visual fraction models and equations. Fractions include: denominators 2, 3, 4, 5, 6, 8, 10, 12, and 100	Mission 5, Topic F, Lesson 31, 33-34
4.CAR.6	Multiply a fraction by a whole number using visual fraction models and equations. Fractions include: denominators 2, 3, 4, 5, 6, 8, 10, 12, and 100	Mission 5, Topic E, Lesson 24-25 Mission 5, Topic G, Lesson 35-38

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
4th Grade Standards		Lessons
4.CAR.7	Solve real-world problems involving multiplicative comparison, using drawings and/or equations with a symbol for the unknown number, and distinguish between multiplicative comparison and additive comparison.	Mission 3, Topic A, Lesson 2-3 Mission 3, Topic B, Lesson 5-6 Mission 3, Topic C, Lesson 9-11 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic G, Lesson 26 Mission 5, Topic G, Lesson 39-40 Mission 7, Topic A, Lesson 4 Mission 7, Topic C, Lesson 14
4.CAR.8	Solve multi-step, real-world problems posed with whole numbers and having whole-number answers, using addition, subtraction, multiplication, and division; include problems in which remainders must be interpreted and represent these problems using equations with symbols standing for the unknown quantity.	Mission 3, Topic D, Lesson 12-13 Mission 3, Topic E, Lesson 14, 19 Mission 3, Topic G, Lesson 31-32
4.CAR.9	Solve real-world problems involving the addition and subtraction of fractions; include mixed numbers with like denominators, using visual fraction models or equations. Fractions include: denominators 2, 3, 4, 5, 6, 8, 10, 12, and 100	Mission 5, Topic D, Lesson 19
4.CAR.10	Solve real-world problems involving the multiplication of a fraction by a whole number using visual fraction models or equations. Fractions include: denominators 2, 3, 4, 5, 6, 8, 10, 12, and 100	Mission 5, Topic G, Lesson 36-40
4.CAR.11	Generate a number or shape pattern that follows a given rule, identifying apparent features of the pattern that are not explicit in the rule itself.	Mission 5, Topic H, Lesson 41
Geometry & Measurement		
4.GM.1	Identify angles as geometric shapes that are formed where two rays share a common endpoint, understanding that angles are measured with reference to a circle so that an angle that turns through a $\frac{1}{360}$ of a circle is called a “one-degree angle” and an angle that turns through n one-degree angles is said to have an angle measure of n degree.	Mission 4, Topic A, Lesson 1 Mission 4, Topic B, Lesson 5-7

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
4th Grade Standards		Lessons
4.GM.2	Measure angles in whole-number degrees, using a protractor, drawing angles of specified measure.	Mission 4, Topic B, Lesson 5-8
4.GM.3	Solve real-world problems finding unknown angle measures, using addition and subtraction when an angle is decomposed into non-overlapping parts.	Mission 4, Topic C, Lesson 9-11
4.GM.4	Identify and draw points, lines, line segments, rays, angles (right, acute, obtuse), and perpendicular and parallel lines, identifying these in quadrilaterals and triangles.	Mission 4, Topic A, Lesson 1-4 Mission 4, Topic D, Lesson 13-15
4.GM.5	Classify two-dimensional figures based on the presence or absence of parallel lines, perpendicular lines, or angles of a specified size, involving quadrilaterals and triangles. Shapes include: quadrilaterals (trapezoid, parallelogram, rectangle, square, rhombus) and triangles (right, acute, obtuse)	Mission 4, Topic D, Lesson 13-15
4.GM.6	Identify and/or draw lines of symmetry for a two-dimensional figure.	Mission 4, Topic D, Lesson 12
4.GM.7	Apply the area and perimeter formulas for rectangles and figures composed of two or more rectangles in real-world situations.	Mission 3, Topic A, Lesson 1-3 Mission 7, Topic D, Lesson 15-16
4.GM.8	Convert measurements of length, weight/mass, and liquid volume within the same system of measurement, metric and customary, expressing measurements from a larger unit in terms of a smaller unit.	Mission 2, Topic A, Lesson 1-3 Mission 2, Topic B, Lesson 4 Mission 4, Topic A, Lesson 4 Mission 7, Topic A, Lesson 1-3 Mission 7, Topic B, Lesson 6-9 Mission 7, Topic C, Lesson 12-13
4.GM.9	Solve real-world problems involving time intervals that may cross the hour.	Mission 7, Topic A, Lesson 3-4 Mission 7, Topic B, Lesson 10-11 Mission 7, Topic C, Lesson 14
4.GM.10	Solve real-world problems involving addition and subtraction of money, including the ability to make change.	Mission 6, Topic E, Lesson 15-16

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
4th Grade Standards		Lessons
4.GM.11	Solve real-world problems involving distances, liquid volume, and masses of objects, including problems that require expressing measurements given in a larger unit in terms of a smaller unit.	Mission 2, Topic A, Lesson 1-3 Mission 2, Topic B, Lesson 5 Mission 5, Topic G, Lesson 40 Mission 6, Topic D, Lesson 14 Mission 7, Topic A, Lesson 1-5 Mission 7, Topic B, Lesson 6-8, 10-11 Mission 7, Topic C, Lesson 14
Data Analysis		
4.DA.1	Collect and interpret data from observations, surveys, and experiments; represent data using frequency tables and scaled bar graphs.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
4.DA.2	Use a line plot to display a data set of measurements in fractions of a unit, solving problems involving addition and subtraction of fractions with like denominators using data presented in line plots.	Mission 5, Topic E, Lesson 28 Mission 5, Topic G, Lesson 40

5th Grade

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
5th Grade Standards		Lessons
Number & Place Value		
5.NPV.1	Recognize that, in a multi-digit number, a digit in a given place represents 10 times as much as it represents in the place to its right and $\frac{1}{10}$ of what it represents in the place to its left.	Mission 1, Topic A, Lesson 1-4 Mission 2, Topic A, Lesson 2 Mission 2, Topic E, Lesson 16
5.NPV.2	Explain patterns in the number of zeros and/or the decimal point when multiplying or dividing a number by a power of 10, using whole-number exponents to denote powers of 10.	Mission 1, Topic A, Lesson 1-4 Mission 2, Topic A, Lesson 1 Mission 2, Topic C, Lesson 10-12 Mission 2, Topic D, Lesson 13-14 Mission 2, Topic G, Lesson 24
5.NPV.3	Read and write decimals to thousandths, using base-ten numerals, word form, and a variety of expanded forms.	Mission 1, Topic B, Lesson 5-6 Mission 1, Topic D, Lesson 9-10 Mission 1, Topic E, Lesson 11-12 Mission 1, Topic F, Lesson 13-15
5.NPV.4	Apply place value understanding to round decimals to any place up to the thousandths.	Mission 1, Topic C, Lesson 7-8
5.NPV.5	Compare two decimals to thousandths based on the value of the digits in each place, using symbols ($<$, $=$, $>$) to record the results of comparisons.	Mission 1, Topic B, Lesson 6
5.NPV.6	Use visual models to explain the product of multiplying a whole number by a fraction greater than and less than one.	Mission 4, Topic F, Lesson 22-23
Computation & Algebraic Reasoning		
5.CAR.1	Use computational fluency to multiply multi-digit whole numbers by using strategies and algorithms, including the standard algorithm, with mastery by the end of fifth grade.	Mission 2, Topic B, Lesson 5-9 Mission 2, Topic C, Lesson 10-12 Mission 2, Topic F, Lesson 20-23
5.CAR.2	Calculate whole number quotients of whole numbers with up to four-digit dividends and two-digit divisors using strategies based on place value, properties of	Mission 2, Topic E, Lesson 16-18 Mission 2, Topic F, Lesson 19-23 Mission 2, Topic H, Lesson 28-29

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
5th Grade Standards		Lessons
	operations, divisibility rules, and the relationship between multiplication and division.	Mission 4, Topic G, Lesson 30-31
5.CAR.3	Add and subtract decimals to the hundredths using concrete models or drawings and strategies based on place value, properties of operations, or the relationship between addition and subtraction.	Mission 1, Topic D, Lesson 9-10
5.CAR.4	Multiply and divide decimals to hundredths, using concrete models or drawings and strategies based on place value, properties of operations, or the relationship between multiplication and division.	Mission 1, Topic E, Lesson 11-12 Mission 1, Topic F, Lesson 13-16 Mission 2, Topic C, Lesson 10-12 Mission 2, Topic D, Lesson 13-14 Mission 2, Topic G, Lesson 24-27 Mission 2, Topic H, Lesson 28-29 Mission 4, Topic E, Lesson 17-18 Mission 4, Topic G, Lesson 29-31
5.CAR.5	Add and subtract fractions with like and unlike denominators by using equivalent fractions $\{a/b = (n \cdot a)/(n \cdot b)\}$ to create common denominators; include real-world problems. Fractions include: mixed numbers	Mission 3, Topic A, Lesson 1-2 Mission 3, Topic B, Lesson 3-6 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 14
5.CAR.6	Interpret and solve fractions as division problems, $(a/b = a \div b)$, where a and b are natural numbers.	Mission 4, Topic B, Lesson 2-5
5.CAR.7	Use visual models and equations to multiply whole numbers by fractions and fractions by fractions, including mixed numbers and fractions greater than one.	Mission 4, Topic C, Lesson 8-9 Mission 4, Topic E, Lesson 13-15, 17-18
5.CAR.8	Apply previous understanding of division to divide unit fractions by whole numbers and whole numbers by unit fractions.	Mission 4, Topic G, Lesson 25-31, 33
5.CAR.9	Solve and create real-world problems involving multiplication of fractions and mixed numbers.	Mission 4, Topic C, Lesson 6-7 Mission 4, Topic D, Lesson 10-12 Mission 4, Topic E, Lesson 13-18 Mission 4, Topic F, Lesson 22-24 Mission 5, Topic C, Lesson 12-15
5.CAR.10	Solve real-world problems involving the division of natural numbers leading to answers in the form of	Mission 4, Topic B, Lesson 3-5

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
5th Grade Standards		Lessons
	fractions or mixed numbers using visual models and equations.	
5.CAR.11	Solve real-world problems involving the division of unit fractions by whole numbers and whole numbers by unit fractions, using visual fraction models and equations.	Mission 4, Topic G, Lesson 25-28, 33
5.CAR.12	Evaluate numerical expressions with parentheses or brackets and exponents with the base of ten, using the Order of Operations.	Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 3-5 Mission 4, Topic D, Lesson 10 Mission 4, Topic H, Lesson 32
5.CAR.13	Write simple expressions that record calculations with numbers, interpreting numerical expressions without evaluating them.	Mission 2, Topic B, Lesson 3-4 Mission 4, Topic D, Lesson 10 Mission 4, Topic H, Lesson 32
5.CAR.14	Generate two numerical patterns given two rules, identifying the relationship between the corresponding terms by graphing the terms in the first quadrant of the coordinate grid.	Mission 6, Topic B, Lesson 7
Geometry & Measurement		
5.GM.1	Classify two-dimensional figures in a hierarchy based on properties with the focus on quadrilaterals and triangles when teaching hierarchies. Shapes to include: quadrilaterals (trapezoid, parallelogram, rectangle, square, rhombus, kite) and triangles (right, acute, obtuse, scalene, isosceles, equilateral)	Mission 5, Topic D, Lesson 16-19, 21
5.GM.2	Find the area of a rectangle with fractional and/or mixed number side lengths by using models and multiplying the fractional side lengths showing that both strategies produce the same area.	Mission 4, Topic E, Lesson 13-15 Mission 5, Topic C, Lesson 10-15
5.GM.3	Measure volumes by counting unit cubes using cubic cm (cm^3), cubic in (in^3), cubic ft (ft^3), and improvised units (u^3).	Mission 5, Topic A, Lesson 1-3
5.GM.4	Solve real-world and mathematical problems involving the volume of rectangular prisms with whole	Mission 5, Topic B, Lesson 6-9

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
5th Grade Standards		Lessons
	number side lengths by applying the formulas ($V = l \cdot w \cdot h$ or $V = B \cdot h$) and the properties of operations.	
5.GM.5	Solve real-world problems by calculating volumes of solid figures composed of two non-overlapping right rectangular prisms by adding the volumes of the non-overlapping parts.	Mission 5, Topic B, Lesson 6, 8-9
5.GM.6	Convert among different-sized standard measurement units within the same system, including both the metric and customary systems, and solve multi-step, real-world problems using conversions.	Mission 1, Topic A, Lesson 4 Mission 2, Topic D, Lesson 13-15 Mission 4, Topic C, Lesson 9 Mission 4, Topic E, Lesson 19-20 Mission 5, Topic B, Lesson 5
5.GM.7	Graph points with whole number coordinates on a coordinate plane in the first quadrant, explaining how the coordinates relate to the horizontal and vertical axes to describe the location of points in the plane.	Mission 6, Topic A, Lesson 1-6 Mission 6, Topic B, Lesson 7-12 Mission 6, Topic C, Lesson 13-17 Mission 6, Topic D, Lesson 18, 20 <i>Zearn Math includes lessons on graphing points on a coordinate plane extending to fractional numbers. Teachers might need to adjust their instruction to ensure proper alignment.</i>
5.GM.8	Represent real-world and mathematical problems by graphing points in the first quadrant on a coordinate plane, interpreting coordinate values of points in the context of the situation.	Mission 6, Topic C, Lesson 13-17 Mission 6, Topic D, Lesson 18-20
Data Analysis		
5.DA.1	Collect and interpret data from observations, surveys, and experiments; represent data using frequency tables, scaled bar graphs, and scaled line graphs.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
5.DA.2	Use a line plot to display a data set of measurements in fractions of a unit solving problems involving all four operations with fractions (excluding division of a fraction by fraction) using data presented in line plots.	Mission 4, Topic A, Lesson 1 Mission 4, Topic D, Lesson 10

6th Grade

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
6th Grade Standards		Lessons
Number Concepts & Computations		
6.NCC.1	Explain positive and negative integers as being opposite values or directions and the meaning of 0 in a real-world context.	Mission 7, Topic A, Lesson 1, 5
6.NCC.2	Find and plot rational numbers on horizontal and vertical number lines in real-world and mathematical problems.	Mission 7, Topic A, Lesson 2 Mission 7, Topic C, Lesson 11-13, 15
6.NCC.3	Compare rational numbers, using inequalities ($<$, $>$, $\leq$, $\geq$, $\neq$) and order on a number line.	Mission 7, Topic B, Lesson 8-10
6.NCC.4	Interpret the absolute value of numbers for positive or negative quantities in a real-world context.	Mission 7, Topic A, Lesson 6 Mission 7, Topic C, Lesson 13
6.NCC.5	Convert between fractions, decimals, and percents in real-world and mathematical problems.	Mission 3, Topic D, Lesson 10-16
6.NCC.6	Interpret and represent quotients of fractions. Fractions include all forms of fractions.	Mission 4, Topic A, Lesson 3 Mission 4, Topic B, Lesson 4-9 Mission 4, Topic C, Lesson 10-11 Mission 4, Topic D, Lesson 12-14 Mission 4, Topic E, Lesson 16-17
6.NCC.7	Solve problems involving the division of fractions in real-world and mathematical problems. Fractions include all forms of fractions.	Mission 4, Topic A, Lesson 3 Mission 4, Topic B, Lesson 4-9 Mission 4, Topic C, Lesson 10-11 Mission 4, Topic D, Lesson 12-14 Mission 4, Topic E, Lesson 16-17
6.NCC.8	Divide multi-digit numbers fluently in real-world and mathematical problems.	Mission 5, Topic D, Lesson 9-11
6.NCC.9	Use any standard algorithm to fluently add and subtract multi-digit decimals and fractions in real-world and mathematical problems.	Mission 5, Topic B, Lesson 2-4 Mission 5, Topic E, Lesson 14-15 Mission 6, Topic A, Lesson 4

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
6th Grade Standards		Lessons
6.NCC.10	Use any standard algorithm to fluently multiply and divide multi-digit decimals and fractions in real-world and mathematical problems.	Mission 5, Topic C, Lesson 7-8 Mission 5, Topic D, Lesson 12-13 Mission 5, Topic E, Lesson 14-15 Mission 8, Topic C, Lesson 12 Mission 9, Topic A, Lesson 6
6.NCC.11	Solve real-world and mathematical problems with the greatest common factor of two whole numbers less than or equal to 100.	Mission 7, Topic D, Lesson 16-18
6.NCC.12	Solve real-world and mathematical problems with the least common multiple of two whole numbers less than or equal to 12.	Mission 7, Topic D, Lesson 16-18
6.NCC.13	Use the distributive property and the greatest common factor to rewrite the sum of two whole numbers, 1 through 100.	Mission 7, Topic D, Lesson 16-18
Proportional Relationships		
6.PR.1	Use precise ratio language and notation to describe a ratio as a relationship between two quantities.	Mission 2, Topic A, Lesson 1-2 Mission 2, Topic B, Lesson 3-5 Mission 9, Topic A, Lesson 4
6.PR.2	Calculate unit rates to include unit pricing and constant speed.	Mission 2, Topic C, Lesson 8-10 Mission 3, Topic C, Lesson 5-9 Mission 6, Topic D, Lesson 16 Mission 6, Topic E, Lesson 17
6.PR.3	Give examples of unit rates as a ratio that compares two quantities with different units of measure, limited to non-complex fractions.	Mission 2, Topic C, Lesson 10 Mission 3, Topic A, Lesson 1 Mission 3, Topic C, Lesson 5-7 Mission 9, Topic A, Lesson 6
6.PR.4	Create various representations to compare ratios and find missing values to solve real-world and mathematical problems.	Mission 2, Topic C, Lesson 6-10 Mission 2, Topic D, Lesson 11-14 Mission 2, Topic E, Lesson 15-16 Mission 2, Topic F, Lesson 17 Mission 3, Topic B, Lesson 3-4 Mission 3, Topic C, Lesson 5-9 Mission 3, Topic D, Lesson 10-15 Mission 6, Topic D, Lesson 16 Mission 6, Topic E, Lesson 17

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
6th Grade Standards		Lessons
		Mission 9, Topic A, Lesson 4-6
6.PR.5	Find a percent of a quantity as a rate per 100 and solve problems involving finding the whole when given a part and the percent.	Mission 3, Topic D, Lesson 10-16 Mission 6, Topic B, Lesson 7 Mission 9, Topic A, Lesson 4, 6
Algebra		
6.ALG.1	Read and write expressions in real-world or mathematical problems in which letters stand for numbers.	Mission 6, Topic B, Lesson 10-11
6.ALG.2	Use mathematical terms to identify parts of an expression, including the names of operations, terms, factors, coefficients, variables, and constants.	Mission 7, Topic B, Lesson 10
6.ALG.3	Write and evaluate expressions for given values of variables, using order of operations, including expressions with whole number exponents.	Mission 1, Topic B, Lesson 5-6 Mission 1, Topic C, Lesson 9-10 Mission 6, Topic B, Lesson 6 Mission 6, Topic C, Lesson 14-15
6.ALG.4	Generate equivalent expressions by applying the associative, commutative, distributive, and identity properties.	Mission 6, Topic B, Lesson 10-11
6.ALG.5	Identify when two expressions are equivalent by using properties of operations including like terms.	Mission 5, Topic D, Lesson 13 Mission 6, Topic B, Lesson 8, 10-11
6.ALG.6	Use substitution to determine if a given value in a specified set makes an equation or inequality true. Include the following inequality symbols: $<$, $>$, $\leq$, $\geq$, $\neq$	Mission 6, Topic A, Lesson 2-5 Mission 6, Topic B, Lesson 8 Mission 6, Topic C, Lesson 15 Mission 7, Topic B, Lesson 9-10
6.ALG.7	Write and solve one-step equations in real-world and mathematical problems, involving positive rational numbers and zero.	Mission 6, Topic A, Lesson 3-5 Mission 6, Topic B, Lesson 7
6.ALG.8	Write, solve, and graph one-step inequalities in real-world and mathematical problems.	Mission 7, Topic B, Lesson 8-10
Geometry & Measurement		

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
6th Grade Standards		Lessons
6.GM.1	Find the area of triangles, quadrilaterals, and polygons by composing or decomposing to solve real-world and mathematical problems.	Mission 1, Topic A, Lesson 2-3 Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 7-10 Mission 1, Topic D, Lesson 11 Mission 1, Topic G, Lesson 19 Mission 4, Topic D, Lesson 14
6.GM.2	Apply the formulas $V = lwh$ and $V = Bh$ to find the volume of right rectangular prisms with fractional edge lengths to solve real-world and mathematical problems, including solving for an unknown dimension.	Mission 1, Topic E, Lesson 15 Mission 4, Topic D, Lesson 14-15 Mission 4, Topic E, Lesson 17
6.GM.3	Construct nets of a rectangular prism, rectangular pyramid, triangular prism, and triangular pyramid, using the nets to find the surface area of these prisms.	Mission 1, Topic E, Lesson 12-16 Mission 1, Topic F, Lesson 18 Mission 1, Topic G, Lesson 19
6.GM.4	Find and graph pairs of rational numbers in all four quadrants of the coordinate plane in real-world and mathematical problems.	Mission 7, Topic C, Lesson 11-13, 15
6.GM.5	Draw polygons in the coordinate plane when given coordinates for the vertices.	Mission 7, Topic C, Lesson 15
6.GM.6	Use coordinates to calculate vertical and horizontal distances between points with the same x-coordinate or the same y-coordinate to solve real-world and mathematical problems.	Mission 7, Topic C, Lesson 15
6.GM.7	Convert measurements within and between the metric and customary measurement systems to solve real-world and mathematical problems.	Mission 3, Topic B, Lesson 3-4 Mission 3, Topic C, Lesson 9
Statistics & Probability		
6.SP.1	Identify the difference between statistical and non-statistical questions and write simple statistical questions that allow variable responses.	Mission 8, Topic A, Lesson 2 Mission 8, Topic B, Lesson 3, 6-7 Mission 8, Topic D, Lesson 17
6.SP.2	Calculate and interpret any measure of center (mean, median, and mode) of a numerical data set.	Mission 8, Topic C, Lesson 9-12 Mission 8, Topic D, Lesson 13-16 Mission 8, Topic E, Lesson 18

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
6th Grade Standards		Lessons
6.SP.3	Determine which measure of center (mean or median) is more appropriate to describe the center of data and justify the choice.	Mission 8, Topic C, Lesson 12 Mission 8, Topic D, Lesson 14-16 Mission 8, Topic E, Lesson 18
6.SP.4	Describe how the mean or median is affected by outliers of a numerical data set.	Mission 8, Topic D, Lesson 13-14
6.SP.5	Calculate and interpret the measure of variation [range and interquartile range (IQR)] of a numerical data set.	Mission 8, Topic D, Lesson 17
6.SP.6	Determine which measure of variation (range or interquartile range) is more appropriate to describe the shape; justify the choice.	Mission 8, Topic D, Lesson 17
6.SP.7	Represent numerical data on a number line, histogram, and box plot.	Mission 8, Topic B, Lesson 3-8
6.SP.8	Calculate the relative frequency of an interval of data values when given a histogram.	Mission 8, Topic B, Lesson 6-8
6.SP.9	Interpret a box plot to answer statistical questions about a data set.	Mission 8, Topic D, Lesson 16-17

7th Grade

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
7th Grade Standards		Lessons
Number Concepts & Computations		
7.NCC.1	Represent addition and subtraction of rational numbers in real-world contexts using a variety of forms.	Mission 5, Topic A, Lesson 1 Mission 5, Topic A, Lesson 1 Mission 5, Topic A, Lesson 1 Mission 5, Topic B, Lesson 2-3 Mission 5, Topic B, Lesson 2-3 Mission 5, Topic B, Lesson 3 Mission 5, Topic B, Lesson 3, 5-7 Mission 5, Topic B, Lesson 4, 6 Mission 6, Topic D, Lesson 18 Mission 6, Topic D, Lesson 18 Mission 7, Topic B, Lesson 6
7.NCC.2	Model and describe additive inverse in real-world situations to show opposite quantities combine to make 0.	Mission 5, Topic A, Lesson 1 Mission 5, Topic B, Lesson 2-3
7.NCC.3	Demonstrate in real-world contexts the distance between two rational numbers on the number line as the absolute value of their differences.	Mission 5, Topic A, Lesson 1 Mission 5, Topic B, Lesson 3, 5, 7 Mission 6, Topic D, Lesson 18
7.NCC.4	Convert a rational number in fraction form to decimal form and recognize that the decimal form of a rational number terminates in 0s or eventually repeats.	Mission 5, Topic B, Lesson 3
7.NCC.5	Interpret the products and quotients of rational numbers by describing real-world contexts.	Mission 5, Topic C, Lesson 8-9, 11
7.NCC.6	Apply properties of operations as strategies to fluently add, subtract, multiply, and divide rational numbers.	Mission 5, Topic B, Lesson 3 Mission 5, Topic C, Lesson 10-11
7.NCC.7	Use addition and subtraction with rational numbers in any form to solve multi-step problems in real-world and mathematical contexts.	Mission 5, Topic B, Lesson 7 Mission 5, Topic D, Lesson 13-14 Mission 5, Topic E, Lesson 15-16 Mission 5, Topic F, Lesson 17 Mission 9, Topic A, Lesson 3 Mission 9, Topic B, Lesson 6

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
7th Grade Standards		Lessons
7.NCC.8	Use multiplication and division with rational numbers in any form to solve multi-step problems in real-world and mathematical contexts.	Mission 5, Topic C, Lesson 12 Mission 5, Topic D, Lesson 13-14 Mission 5, Topic E, Lesson 15-16 Mission 5, Topic F, Lesson 17 Mission 9, Topic A, Lesson 3 Mission 9, Topic B, Lesson 6
7.NCC.9	Apply operations with rational numbers involving the order of operations, involving nested grouping symbols.	Mission 5, Topic B, Lesson 7 Mission 5, Topic D, Lesson 13-14 Mission 5, Topic E, Lesson 15-16 Mission 5, Topic F, Lesson 17 Mission 9, Topic A, Lesson 3 Mission 9, Topic B, Lesson 6
Proportional Relationships		
7.PR.1	Determine the unit rate (constant of proportionality) from tables, graphs, equations, diagrams, or verbal descriptions of proportional relationships.	Mission 2, Topic A, Lesson 2-3 Mission 2, Topic B, Lesson 5
7.PR.2	Calculate unit rates in real-world contexts that include complex fractions.	Mission 2, Topic C, Lesson 8 Mission 4, Topic A, Lesson 2-3 Mission 9, Topic B, Lesson 5
7.PR.3	Solve multi-step ratio and percent problems in a real-world context, including percent error and percent increase and decrease.	Mission 3, Topic A, Lesson 5 Mission 4, Topic B, Lesson 6-9 Mission 4, Topic C, Lesson 10-15 Mission 4, Topic D, Lesson 16 Mission 9, Topic A, Lesson 1-4 Mission 9, Topic B, Lesson 6, 8 Mission 9, Topic C, Lesson 13
7.PR.4	Determine whether two quantities represent proportional relationships by using equivalent ratios in a table and by graphing on a coordinate plane.	Mission 2, Topic A, Lesson 2-3 Mission 2, Topic D, Lesson 10 Mission 3, Topic A, Lesson 1-3
7.PR.5	Compare two different proportional relationships represented in different forms.	Grade 8, Mission 3, Topic A, Lesson 2-4 Grade 8, Mission 3, Topic B, Lesson 6
7.PR.6	Create equations in the form of $y = mx$ from tables, verbal descriptions, or graphs.	Mission 2, Topic B, Lesson 4-6 Mission 3, Topic A, Lesson 5

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
7th Grade Standards		Lessons
7.PR.7	Given a graph with a proportional relationship, explain the meaning of a point (x, y) on the graph, including the origin $(0, 0)$ and the unit rate $(1, r)$.	Mission 2, Topic D, Lesson 11
Algebra		
7.ALG.1	Generate and justify equivalent expressions, using properties of operations to add, subtract, factor, and expand linear expressions with rational coefficients within mathematical and real-world problems.	Mission 6, Topic D, Lesson 18-22 Mission 9, Topic B, Lesson 7
7.ALG.2	Model and solve fluently two-step equations in real-world or mathematical problems.	Mission 5, Topic E, Lesson 15-16 Mission 6, Topic A, Lesson 4-6 Mission 6, Topic B, Lesson 7-12 Mission 9, Topic B, Lesson 7
7.ALG.3	Create, solve, and graph two-step inequalities in real-world and mathematical problems in the forms $px \pm q > r$, $px \pm q < r$, $px \pm q \geq r$, and $px \pm q \leq r$.	Mission 6, Topic C, Lesson 14, 16-17
7.ALG.4	Write an equation to express two quantities in terms of the dependent and independent variables.	Grade 6, Mission 6, Topic D, Lesson 16-18
7.ALG.5	Describe the relationship between the dependent and independent variables in an equation using tables and graphs, relating these to the equation.	Grade 6, Mission 6, Topic D, Lesson 16-18
Geometry & Measurement		
7.GM.1	Describe the proportional relationship between the circumference and diameter of a circle.	Mission 3, Topic A, Lesson 3-5 Mission 3, Topic B, Lesson 7-9 Mission 3, Topic C, Lesson 10-11 Mission 9, Topic A, Lesson 4 Mission 9, Topic C, Lesson 11-12
7.GM.2	Use area and circumference formulas of a circle to solve real-world and mathematical problems.	Mission 3, Topic A, Lesson 3-5 Mission 3, Topic B, Lesson 7-9 Mission 3, Topic C, Lesson 10-11 Mission 9, Topic A, Lesson 4 Mission 9, Topic C, Lesson 11-12

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
7th Grade Standards		Lessons
7.GM.3	Apply the formulas for the volume and surface area of right rectangular prisms, rectangular pyramids, triangular prisms, and triangular pyramids to solve real-world and mathematical problems.	Mission 1, Topic A, Lesson 6 Mission 2, Topic C, Lesson 8 Mission 3, Topic B, Lesson 6 Mission 7, Topic C, Lesson 12-16 Mission 7, Topic D, Lesson 17 Mission 9, Topic A, Lesson 4 Mission 9, Topic B, Lesson 5, 9
7.GM.4	Describe the two-dimensional figure that results from slicing a three-dimensional figure parallel and perpendicular to the base. Three-dimensional figures include: right rectangular prisms, triangular prisms, and cylinders	Mission 7, Topic C, Lesson 11, 13
7.GM.5	Solve multi-step problems involving supplementary, complementary, vertical, and adjacent angles to include solving for an unknown angle in a figure.	Mission 7, Topic A, Lesson 2-5
7.GM.6	Calculate the scale factor, compute the actual lengths from the scale in a drawing, and reproduce a scale drawing using another scale.	Mission 1, Topic A, Lesson 1-6 Mission 1, Topic B, Lesson 7-12 Mission 1, Topic C, Lesson 13 Mission 2, Topic A, Lesson 1 Mission 3, Topic B, Lesson 6 Mission 3, Topic C, Lesson 11 Mission 9, Topic A, Lesson 4 Mission 9, Topic C, Lesson 13
Statistics & Probability		
7.SP.1	Interpret data displayed in a histogram and box plot to answer questions about the data.	Mission 8, Topic C, Lesson 11 Mission 8, Topic D, Lesson 18
7.SP.2	Recognize, create, and interpret categorical data in a circle graph.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
7.SP.3	Graph two numerical data sets and compare their variability. Variability includes: range, interquartile range, or mean absolute deviation	Mission 8, Topic C, Lesson 11 Mission 8, Topic D, Lesson 18

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
7th Grade Standards		Lessons
7.SP.4	Select an appropriate measure(s) of center or variability and draw valid comparative inferences for two data sets.	Mission 8, Topic D, Lesson 15-16, 18-19 Mission 8, Topic E, Lesson 20 Mission 9, Topic A, Lesson 3
7.SP.5	Distinguish between a random and non-random sample.	Mission 8, Topic C, Lesson 12-14 Mission 8, Topic D, Lesson 15 Mission 8, Topic E, Lesson 20
7.SP.6	Use a random sampling of a population to draw valid inferences and generalizations of populations.	Mission 8, Topic C, Lesson 13-14 Mission 8, Topic D, Lesson 15-17 Mission 8, Topic E, Lesson 20
7.SP.7	Determine the sample space of a simple experiment and use the sample space to determine the theoretical probability of a given set of outcomes. Simple experiments include: tossing a fair coin, spinning a fair spinner, rolling a fair dice, picking a random marble from a bag, and selecting a random card from a deck	Mission 8, Topic A, Lesson 3-6 Mission 8, Topic C, Lesson 14 Mission 8, Topic E, Lesson 20
7.SP.8	Recognize that probabilities in a simple experiment can be qualitative descriptors of likelihood: impossible (0), unlikely, neither likely nor unlikely, likely, or certain (1).	Mission 8, Topic A, Lesson 2-6
7.SP.9	Determine experimental probabilities in simple experiments and represent as fractions, decimals, and percents.	Mission 8, Topic A, Lesson 4-6
7.SP.10	Use theoretical probability of an event in a simple experiment to predict the number of times that an event will occur for a large number of experiments.	Mission 8, Topic A, Lesson 3 Mission 8, Topic E, Lesson 20

8th Grade

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
8th Grade Standards		Lessons
Number Concepts & Computations		
8.NCC.1	Describe relationships in the real number system (rational and irrational). Numbers relationships to include: decimal expansion for rational and irrational numbers, square roots of nonperfect squares, and cube roots of nonperfect cubes	Mission 8, Topic E, Lesson 14-15
8.NCC.2	Compare the size of irrational numbers and locate them on a number line by finding the rational approximations.	Mission 8, Topic A, Lesson 1 Mission 8, Topic B, Lesson 4-5 Mission 8, Topic D, Lesson 12-13
8.NCC.3	Know and apply the properties of integer exponents to generate equivalent numerical expressions.	Mission 7, Topic B, Lesson 2-8 Mission 7, Topic C, Lesson 11, 14
8.NCC.4	Write very large and very small numbers in scientific notation using positive and negative exponents.	Mission 7, Topic C, Lesson 9-12, 14 Mission 7, Topic D, Lesson 16
8.NCC.5	Compare numbers written in scientific notation to determine how many times larger or smaller one number is than the other, using real-world and mathematical problems.	Mission 7, Topic C, Lesson 9-12, 14 Mission 7, Topic D, Lesson 16
8.NCC.6	Solve real-world and mathematical problems by performing operations with numbers written in standard and scientific notation.	Mission 7, Topic C, Lesson 10-15 Mission 7, Topic D, Lesson 16
8.NCC.7	Solve equations in the form of $x^2 = p$ or $x^3 = p$ where p is a positive rational number.	Mission 8, Topic B, Lesson 2-5 Mission 8, Topic C, Lesson 10 Mission 8, Topic D, Lesson 12-13
8.NCC.8	Evaluate square roots of perfect squares and cube roots of perfect cubes.	Mission 8, Topic D, Lesson 12-13
Functions		

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
8th Grade Standards		Lessons
8.FN.1	Graph proportional relationships, interpreting the unit rate as the slope of the graph.	Mission 3, Topic A, Lesson 2-3 Mission 3, Topic B, Lesson 6
8.FN.2	Explain, using similar right triangles, how the slope of a line is the same between two points on a non-vertical line or non-horizontal line. Slope includes: positive, negative, horizontal (zero), and vertical (undefined)	Mission 2, Topic C, Lesson 10-12 Mission 3, Topic B, Lesson 7 Mission 3, Topic C, Lesson 10-11 Mission 3, Topic E, Lesson 14
8.FN.3	Determine whether a relation is a function or not when given a function map, table, graph, equation, or set of ordered pairs.	Mission 5, Topic A, Lesson 1-2 Mission 5, Topic B, Lesson 3-5 Mission 5, Topic E, Lesson 17 Mission 9, Topic A, Lesson 1
8.FN.4	Compare the rate of change (slope) and y-intercept (initial value) of two linear functions each represented in different forms. Functions are represented algebraically, graphically, numerically in tables, or by verbal descriptions.	Mission 5, Topic B, Lesson 7 Mission 5, Topic C, Lesson 8
8.FN.5	Distinguish between linear and nonlinear functions by comparing graphs and equations.	Mission 5, Topic B, Lesson 4, 7 Mission 5, Topic C, Lesson 8 Mission 5, Topic E, Lesson 18
8.FN.6	Determine the rate of change (slope) and y-intercept (initial value) from tables, graphs, equations, and verbal descriptions of linear relationships.	Mission 5, Topic C, Lesson 8-10 Mission 5, Topic D, Lesson 11
8.FN.7	Interpret and explain the meaning of the rate of change (slope) and y-intercept (initial value) of a linear relationship in a real-world context.	Mission 5, Topic C, Lesson 8-10
8.FN.8	Analyze a graph by describing the functional relationships between two quantities.	Mission 5, Topic B, Lesson 5-6 Mission 5, Topic C, Lesson 10 Mission 5, Topic D, Lesson 11
8.FN.9	Sketch a graph that exhibits qualitative features of a function described verbally.	Mission 5, Topic B, Lesson 5-6
Algebra		

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
8th Grade Standards		Lessons
8.ALG.1	Analyze and solve one-variable linear equations with rational coefficients containing solutions with one, zero, or infinitely many solutions.	Mission 4, Topic B, Lesson 3-9
8.ALG.2	Analyze and solve one-variable linear inequalities with rational coefficients.	Mission 4, Topic B, Lesson 3-9
8.ALG.3	Analyze and solve systems of linear equations in the form $y = mx + b$ in real-world or mathematical contexts, graphically and algebraically.	Mission 3, Topic C, Lesson 13-14 Mission 4, Topic B, Lesson 9 Mission 4, Topic C, Lesson 10-15 Mission 4, Topic D, Lesson 16
Geometry & Measurement		
8.GM.1	Apply the formulas for the volume and surface area of cylinders, cones, and spheres to solve real-world and mathematical problems.	Mission 5, Topic D, Lesson 13-16 Mission 5, Topic E, Lesson 17-21 Mission 5, Topic F, Lesson 22
8.GM.2	Describe the two-dimensional figure that results from slicing a three-dimensional figure parallel and perpendicular to the base. Three-dimensional figures include: pyramids, cones, and spheres	Grade 7, Mission 7, Topic C, Lesson 11, 13
8.GM.3	Model or explain an informal proof of the Pythagorean Theorem and its converse.	Mission 8, Topic C, Lesson 7, 9
8.GM.4	Apply the Pythagorean Theorem to determine unknown side lengths in right triangles.	Mission 8, Topic C, Lesson 6-8
8.GM.5	Apply the Pythagorean Theorem to find the distance between two points in a coordinate system.	Mission 8, Topic C, Lesson 11
8.GM.6	Given a figure, draw a congruent figure on a coordinate plane resulting from a rotation, reflection, or translation.	Mission 1, Topic A, Lesson 2-4, 6 Mission 1, Topic B, Lesson 7-10 Mission 1, Topic C, Lesson 11, 13 Mission 1, Topic D, Lesson 14 Mission 3, Topic B, Lesson 8
8.GM.7	Identify a single transformation used to transform one figure onto another on a coordinate plane. Rotations include: 90°, 180°, and 270°	Mission 1, Topic A, Lesson 5-6 Mission 2, Topic A, Lesson 4-5 Mission 2, Topic C, Lesson 12

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
8th Grade Standards		Lessons
8.GM.8	Given two congruent figures, describe a sequence of transformations that maps one figure to another.	Mission 1, Topic C, Lesson 11-13 Mission 1, Topic D, Lesson 15 Mission 2, Topic B, Lesson 6-7
8.GM.9	Perform a given sequence of transformations of a figure on the coordinate plane, including rotations, reflections, translations, and dilations. Rotations include: 90°, 180°, and 270°	Mission 1, Topic C, Lesson 11-13 Mission 1, Topic D, Lesson 15 Mission 2, Topic B, Lesson 6-7
8.GM.10	Describe the effects of rotations, reflections, translations, and dilations on two-dimensional figures using coordinates. Rotations include: 90°, 180°, and 270°	Mission 1, Topic A, Lesson 5-6 Mission 2, Topic A, Lesson 4-5 Mission 2, Topic C, Lesson 12
8.GM.11	Given two similar two-dimensional figures, describe a sequence of transformations that exhibits similarity, including rotations, reflections, translations, and dilations.	Mission 2, Topic B, Lesson 6-7, 9
Statistics & Probability		
8.SP.1	Construct scatter plots using bivariate data; determine if the data displays a linear or nonlinear pattern and positive, negative, or no association.	Mission 6, Topic A, Lesson 1-2 Mission 6, Topic B, Lesson 3-8
8.SP.2	Construct straight lines to approximately fit data displaying a linear association when presented in scatter plots.	Mission 6, Topic B, Lesson 4-6, 8
8.SP.3	Construct and interpret a relative frequency table, using data from two categorical variables collected from the same subject.	Mission 6, Topic C, Lesson 9-10
8.SP.4	Determine the sample space and use the sample space to determine the theoretical probability of a given set of outcomes for compound experiments, using organized lists, tables, or tree diagrams. Compound experiments include a combination of two different experiments.	Grade 7, Mission 8, Topic B, Lesson 8-9
8.SP.5	Determine theoretical and experimental probabilities of compound experiments.	Grade 7, Mission 8, Topic A, Lesson 6

ARKANSAS MATHEMATICS STANDARDS AND COURSES		ZEARN MATH
8th Grade Standards		Lessons
8.SP.6	Use theoretical probability of an event in a compound experiment to predict the number of times that event will occur for a large number of experiments.	Grade 7, Mission 8, Topic B, Lesson 7-10
		Grade 7, Mission 8, Topic A, Lesson 6 Grade 7, Mission 8, Topic B, Lesson 7-10